

EEE 4483

Assignment

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Department - CSE

Section - 1

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Ans. to the problem no. 1

(i) Given, clock frequency $f_{CLK} = 50 \text{ KHz}$

$$T_{CLK} = \frac{1}{50 \times 10^3} = 0.02 \text{ ms}$$

$$\text{slope of ramp} = 1.953 \text{ V/ms}$$

Since the ~~maximum conversion time is the time the ADC takes for conversion when the input is full scale~~
quantization step size is the smallest input change that results in 1 count or 1 clock cycle,

$$\begin{aligned} \text{quantization step} &= \text{ramp slope} \times T_{CLK} \\ &= (1.953 \times 0.02) \text{ V} \\ &= 39.06 \text{ mV} \end{aligned}$$

(ii) Given, the input voltage is 3.91 V

We know, for a single slope ADC,

$$V_{in} = \frac{V_{ref}}{RC} \times n \times T_{CLK}$$

V_{ref} = Reference voltage
 RC , resistance, capacitance

$$\text{Here, } \frac{V_{ref}}{RC} = \text{slope of ramp generator}$$

T_{CLK}
 n = number of counts

$$\text{So, } V_{in} = 1.953 \times n \times 0.02$$

$$\begin{aligned} \text{or, } n &= (1.953 \times 0.02)^{-1} \times 3.91 \\ &= 100.1 \approx 100 \end{aligned}$$

In binary, this can be represented as 011 001 00

(3 bits representation) This is the required digital output.

(iii) The maximum conversion time an ADC takes for conversion when the input is full scale, that is, the counter reaches maximum count.

The maximum count that can be reached for 8 bit counter = $2^8 = 256$.

$$\begin{aligned}\text{So, max conversion time} &= 256 \times T_{\text{CLK}} \\ &= (256 \times 0.02) \text{ ms} \\ &= 5.12 \text{ ms}\end{aligned}$$

Ans. to the Q.no. 2

(i) The integration phase (run-up) time is upto the point when counter reaches maximum value (N -bit) 2^N or 2^N counts.

The run-up time, $t_1 = 40.96 \text{ ms}$

Clock frequency = 100 kHz

We know,

$$t_1 = 2^N \times T_{\text{CLK}} \quad 2^N = n \text{ (number of counts)}$$

$$\text{or, } n = \frac{t_1}{T_{\text{CLK}}}$$

$$\text{or, } n = \frac{40.96 \times 10^{-3}}{\frac{1}{100 \times 10^3}}$$

$$= 4096$$

(ii) The number of bits in the counter is,

$$\log_2(4096) = 12$$

(iii) The de-integration time be t_2

$$t_1 = 40.96 \text{ ms}$$

$$V_{in} (\text{input voltage}) = 2.41 \text{ V}$$

$$V_{ref} (\text{reference "}) = 5 \text{ V}$$

We know,

$$V_{in} \times t_1 = V_{ref} \times t_2$$

$$\text{so, } t_2 = \left(\frac{2.41}{5} \times 40.96 \right) \text{ ms} = 19.74 \text{ ms}$$

(iv) We know,

$$t_2 = n \times T_{CLK} \quad \text{where } n \text{ is the number of counts}$$

$$n = \frac{19.74 \times 10^{-3}}{\frac{1}{100 \times 10^3}} = 1974$$

1974 is the required number and in binary code,
it is 011110110110 (12 bit representation)

This is the required digital output code. corresponding
to 2.41 V.

Ans. to the Q.no-3

(i)

For single slope ADC,

We find $n \propto \frac{V_{in}}{V_{ref}}$

N is bit number

V_{in} = input voltage

V_{ref} = reference "

$$n = \frac{V_{in}}{V_{ref}} \times 2^N = \frac{4.5}{5} \times 2^{12} = 3686.4 \approx 3686$$

(count done)

$$T_{clk} = \frac{1}{10^6} \text{ s}$$

$$\begin{aligned} \text{Conversion time } t_1 &= n \times T_{clk} = 3686.4 \times \frac{1}{10^6} \\ &= 3.6864 \text{ ms} \end{aligned}$$

For dual slope ADC,

$t_1 + t_2$ is the required conversion time where

t_1 is the run-up time for integration and

t_2 is the de-integration time.

$$t_1 = 2^N \times T_{clk} = 4096 \times \frac{1}{10^6} = 4.096 \text{ ms}$$

$$t_2 = \frac{V_{in}}{V_{ref}} \times t_1 = \frac{4.5}{5} \times 4.096 = 3.6864 \text{ ms}$$

$$\begin{aligned} \therefore \text{Conversion time } t_1 + t_2 &= (4.096 + 3.6864) \text{ ms} \\ &= 7.7824 \text{ ms} \end{aligned}$$

(ii) For single slope ADC,

$$\text{Number of counts, } n = \frac{V_{in}}{V_{ref}} \times 4096 = 3686 \text{ (approx.)}$$

For dual slope ADC,

$$n = \frac{4.5}{5} \times \frac{V_{in}}{V_{ref}} \times 2^N = \frac{4.5}{5} \times 4096 = 3686$$

Both gives the same result.

Converting 3686 into binary, we get,

1110 0110 0110 (12 bit representation)

This is the required output code (digital)

(iii) The dual slope ADC is more suitable because,

In case of single slope, we see,

the number of counts, n which is used to determine the digital output code is represented

by -

$$n = \frac{V_i \times RC}{V_{ref} \times T_{CLK}}$$

V_i = Input voltage

RC = Resistance $\times$ Capacitance

T_{CLK} = clock cycle time

V_{ref} = Reference voltage

Here, RC , this value can be prone to fluctuation due to

changes in temperature, environmental factors etc.

Thus, this increases chances of noise disrupting the results.

However, in case of dual slope ADC, this limitation is not present because,

$$n = \frac{V_{in}}{V_{ref}} \times 2^N$$

N = number of bits

Where none of the variables V_{in} and V_{ref} are prone to fluctuations due to previously mentioned factors and 2^N is a constant. That is, n does not depend on anything that can cause noise and might deviate the value in a noisy environment.

Thus, dual slope ADC is more suitable.